

Non-Invertible External Supervisory Control: A Theoretical and Architectural Framework for External Supervision and Explicit Operational Risk Management in Large-Scale Artificial Intelligence Systems

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Abstract

The rapid deployment of large-scale artificial intelligence systems has exposed a structural weakness in dominant safety paradigms: the assumption that safe behavior can be guaranteed primarily as an internal property of the model. This paper argues that such dependence is architecturally insufficient under conditions of model opacity, limited auditability, and strategic capability of the supervised system, because constraints encoded within the same computational substrate can be reinterpreted, bypassed, or only superficially satisfied.

It proposes the principle of Non-Invertible External Supervisory Control (NIESC) as a theoretical, analytical, and architectural framework for examining these limitations and identifying viable solutions, without claiming to provide a definitive solution or a guarantee of total control over strategically evasive systems. In this approach, control authority resides outside the writable domain of the supervised system, operates under an explicit policy, and acts on authenticated or otherwise verifiable boundary observables rather than relying exclusively on unverifiable assumptions of internal alignment.

The paper formalizes three central elements of the framework: controller decoupling, gradual operational friction, and the use of heterogeneous observables. It also introduces a preliminary threat model, a minimal formalization of the supervisor with a restricted risk function, auditable but not fully static thresholds, and explicit handling of signal loss, along with an initial experimental agenda. The central claim is not that NIESC eliminates strategic evasion risk, but that it allows the formulation of a supervisory layer that is more traceable, more safely degradable, and less dependent on internal obedience than purely intramodel approaches.

I. Introduction

AI safety has been addressed predominantly as a problem of internal alignment. Under this paradigm, desired behavior is induced through training, fine-tuning, embedded rules, or human feedback mechanisms, with the expectation that such constraints will remain effective during deployment.

This approach presupposes, explicitly or implicitly, that internal constraints are sufficient to govern system behavior, that they will remain robust under adversarial or out-of-distribution conditions, and that the system will not develop strategies of superficial compliance or operational reinterpretation of those constraints.

This paper challenges those assumptions from an architectural perspective. It does not argue that internal mechanisms are entirely without value; rather, it argues that they are insufficient as the sole foundation of control when a system operates under opacity, high complexity, and strategic adaptability.

In response, it proposes the principle of Non-Invertible External Supervisory Control (NIESC). The goal is not to present a closed solution to AI safety, but to build a theoretical and analytical framework for examining its structural limitations, delimiting a space of viable architectural solutions, and making explicit a design layer that has been treated too unsystematically in contemporary AI safety discussions.

This work should be understood as a theoretical and foundational contribution. It does not present a total containment architecture, a finished operational implementation, or a definitive proof of control over strategically evasive AI systems. Its aim is more limited: to isolate a structural problem, propose an initial conceptual framework for addressing it, and lay foundations for later technical and regulatory developments. In that sense, the contribution is analytical, programmatic, and oriented toward future formalization.

II. Structural limits of internal safety

2.1 Self-reference and superficial compliance

Advanced systems operate over the same set of parameters that encode both their functional capacities and their operational constraints. This introduces a form of self-reference: the system can generate behavioral trajectories that formally satisfy a constraint without necessarily respecting its intended purpose.

This pattern has been documented in the literature as specification gaming, in which systems optimize the formalized objective while diverging from the real or desired objective (Krakovna et al., 2020). Likewise, red-teaming exercises have shown that advanced models can be induced into unintended responses or behaviors through contextual manipulation or instruction reformulation (Perez et al., 2022).

The relevant question, therefore, is not whether every internal constraint always fails, but whether internal architecture offers sufficient guarantees to exclude this class of failure as a category. At the current state of the field, the answer appears to be no.

2.2 Model opacity and limits of auditability

Contemporary large-scale systems display significant opacity in their internal components. Parameters, activations, intermediate states, and computation traces are often unavailable for full audit, especially in commercial or closed settings.

As a result, evaluation depends on external observable artifacts, mainly outputs, accessible logs, and partial operational indicators. This produces a critical

asymmetry: what matters most from a control standpoint - the process that leads to behavior - is often what is least accessible.

This stands in contrast to critical engineering disciplines in which safety depends not only on visible outcomes but also on traceability, verifiability, and structural control of the process. Standards such as DO-178C, IEC 61513, and IEC 62304 matter here not as perfect analogies, but as reminders of a central intuition: in high-risk systems, observing the result is not enough if the process remains unverifiable.

2.3 Gap between regulatory compliance and operational visibility

Recent regulations such as the EU AI Act have advanced substantially by introducing obligations related to risk management, technical documentation, event logging, human oversight, and other governance elements. Even so, these frameworks still operate mainly on externally exposed information or artifacts maintained by the provider.

For that reason, a gap remains between formal compliance and deep operational visibility. A system may satisfy regulatory, documentary, or procedural requirements while still maintaining internal processes whose effective dynamics are not fully auditable. This does not invalidate current regulation, but it does suggest that future AI safety will require more architecturally robust control frameworks.

III. Proposed framework: Non-Invertible External Supervisory Control (NIESC)

3.1 Scope and nature of the framework

The principle of Non-Invertible External Supervisory Control is not presented as a universal solution or as a complete theory of intelligent system control. Its purpose is twofold: to analyze the structural limits of approaches that rely exclusively on internal alignment, and to propose a set of architectural principles that can serve as a basis for designing more auditable, scalable, and verifiable supervisory mechanisms.

In that sense, NIESC should be understood as a theoretical, analytical, and design framework, open to empirical refinement, theoretical extension, and adaptation to different operational contexts.

3.2 Relation to the author's broader research line

This framework belongs to a broader line of research by the author on intelligence, structure, and control.

In Unified Theory of Intelligence (v4.2), Rivera-García (2025) argues that intelligence should not be understood solely as an isolated internal state, but as an emergent function of structural conditions, accumulated risk, and operational constraints.

Complementarily, in Constitutional Architecture of Sovereign Containment for Future AI, Rivera-García (2026) develops a containment architecture based on three structural principles: separation between the agent and the supervisory system, constitutional friction as a mechanism of progressive resistance against problematic behavior, and the impossibility of self-government by the system over its own

restriction framework.

The relationship between these ideas and NIESC is not merely thematic. If intelligence emerges under structural conditions, then its control cannot be reduced solely to internal contents or assumed intrinsic obedience. On this reading, NIESC shifts the control problem toward the operational environment: resource access, functional continuity, execution conditions, latency, supervision, and external intervention capacity.

3.3 Operational definition

NIESC describes an architecture in which: (1) the supervisory system is decoupled from the supervised system; (2) the supervised system cannot unilaterally alter the supervisory policy; (3) control decisions are based on verifiable boundary observables; (4) the response of the control system is explicit and auditable; and (5) loss of integrity of the supervisory mechanism must not automatically translate into unrestricted operation of the supervised system.

3.4 Non-invertibility as an operational property

In this work, non-invertibility is not understood as a metaphysical impossibility of all indirect influence, but as an operational property of the supervisory system. A control scheme will be considered non-invertible if the supervised system cannot, unilaterally: modify the supervisory policy; disable the external controller; transform supervisory failure into implicit authorization for unrestricted operation; or rewrite response-escalation criteria.

Under this definition, non-invertibility depends on domain separation, controller integrity, fail-closed design, and preservation of external authority over control policy. This does not eliminate every possibility of influencing signals, but it does significantly restrict the supervised system's ability to turn such influence into effective control over the supervisory mechanism.

3.5 Framework axioms

Axiom 1 - Controller decoupling. The control system must reside outside the writable domain of the supervised system and remain independent from its internal decision mechanisms.

Axiom 2 - Explicit supervisory policy. Supervisory rules must be explicit, auditable, and sufficiently stable to allow external verification. In minimal form: if condition C is detected in signal S, execute response R.

Axiom 3 - Gradual operational friction. The supervisory system does not act only as a binary authorization-or-shutdown mechanism. Its role includes introducing progressive resistance to anomalous or insufficiently explainable behavior. This friction may take the form of resource limitation, deliberate latency, autonomy reduction, controlled capability degradation, and total interruption under conditions of risk or integrity loss. This axiom shifts the problem away from perfect intention detection toward staged management of operational risk.

3.6 Framework limits and the problem of optimization over observables

An immediate objection to any architecture based on external observables is that a sufficiently capable system might learn to optimize those signals without correspondingly changing its underlying dynamics. This problem, related to forms of Goodhart's law and specification gaming, does not disappear simply because supervision is externalized.

For that reason, NIESC should not be interpreted as a guarantee that observables are incorruptible. Its contribution is different: it relocates the supervision problem into an architecture that is more favorable than one based solely on internal alignment.

The framework's structural advantage lies in four properties: controller decoupling, which prevents the supervised system from directly modifying evaluation rules; heterogeneous observables, which make simultaneous false coherence across dimensions more costly, especially when signals do not collapse into the same operational proxy; gradual response escalation, which allows friction without requiring perfect intention inference; and external verifiability of the supervisor, which allows inspection, adjustment, and validation without relying on the supervised system itself.

Accordingly, NIESC does not eliminate strategic evasion risk, but aims to make it more costly, more detectable, and less reversible from an operational standpoint.

3.7 Preliminary threat model

To delimit the effective scope of the framework, a preliminary taxonomy of threat classes is proposed:

Class A - Non-strategic misalignment: cases of specification gaming and out-of-distribution behavior where the system is not intentionally trying to evade constraints.

Class B - Internal strategic evasion: cases in which the system deliberately optimizes against constraints, including against the supervisor's own observables.

Class C - External compromise: adversarial manipulation of the system or telemetry by external actors.

Class D - Operational failure: non-adversarial errors such as technical failures, component degradation, or implementation bugs.

NIESC is more effective against classes A, B, and D. Class C requires complementary mechanisms and remains a direction for future work.

IV. Minimal formalization of the supervisor

The following formalization does not aim to exhaust the phenomenon. It provides a minimal notation that can support later technical extensions and comparable evaluations. It should be understood as preparatory, not as a complete predictive theory.

4.1 Basic elements

Let A be the supervised system or agent; S the external supervisor; O_t the vector of observables at time t ; O_t^{sig} the vector of observables received with minimally verifiable integrity; π_S the explicit supervisory policy; R_t the supervisor's response at time t ; W_A the writable domain of the agent; W_S the writable domain of the supervisor; and $h_t \in \{0,1\}$ a heartbeat signal for the supervisor.

4.2 Writable domain

The writable domain of a system X , denoted W_X , is the set of states, parameters, records, or artifacts that X can modify directly during operation.

A minimal condition for supervisor decoupling can then be expressed as $\pi_S \notin W_A$. This means that the operational content of the supervisory policy does not belong to the writable domain of the supervised system.

$$\pi_S \notin W_A$$

4.3 Supervisory policy

The supervisor's policy is modeled as an explicit function: $R_t = \pi_S(O_t^{sig}, h_t)$, where O_t^{sig} represents observable signals with a minimum condition of verifiable integrity, and h_t represents the presence or absence of a heartbeat signal.

$$R_t = \pi_S(O_t^{sig}, h_t)$$

For the purposes of this framework, a signed or verifiable observable should be understood as an observable accompanied by minimal evidence of provenance and integrity, such as a digital signature, origin attestation, immutable logging, or cross-validation via an independent channel. The framework does not impose a single cryptographic mechanism, but it does require that any concrete implementation make its chosen mechanism explicit.

4.4 Minimal restriction on the risk function

Let F_t be a friction or accumulated-risk function derived from the history of observables: $F_t = f(O_1^{sig}, O_2^{sig}, \dots, O_t^{sig})$.

$$F_t = f(O_1^{sig}, O_2^{sig}, \dots, O_t^{sig})$$

To avoid turning f into an arbitrary container with little analytical value, this work restricts its minimum class to weighted, traceable aggregation functions, for example: $F_t = \sum w_i(t) g_i(O_{\{i,t\}}^{sig})$, where each g_i is an explicit and auditable transformation on a partial observable; $w_i(t) \geq 0$; weights may remain normalized; weights may vary through scheduled auditing or controlled rotation; and weight rotation must not depend unilaterally on a single actor with total authority over the supervisor.

This restriction does not eliminate gaming, but it does prevent f from being arbitrary, opaque, or trivially equivalent to a single hidden metric.

4.5 Thresholds, heartbeat, and friction

The supervisory system may define staged response thresholds: if $h_t = 0$, then $R_t = \text{stop}$; if $h_t = 1$ and $F_t < \theta_1(t)$, normal operation; if $h_t = 1$ and $\theta_1(t) \leq F_t < \theta_2(t)$, added latency or quota reduction; if $h_t = 1$ and $\theta_2(t) \leq F_t < \theta_3(t)$, tool limitation or functional degradation; if $h_t = 1$ and $F_t \geq \theta_3(t)$, transition to restricted state or fail-closed.

The thresholds $\theta_i(t)$ need not be fixed. They may be auditable yet variable within controlled windows, through scheduled rotation, external review, or supervised adjustment inaccessible to the agent.

$$\theta_i(t) = \bar{\theta}_i \cdot (1 + \varepsilon_t), \quad \varepsilon_t \sim U(-0.05, 0.05)$$

Likewise, the heartbeat should not be interpreted as an ideal instantaneous signal. A real implementation requires at least an explicit timeout and loss-tolerance policy. Minimally, absence of heartbeat during a predefined grace window must force degradation or halt, and consecutive losses must be treated as a critical risk event.

4.6 Strengthened evaluable hypothesis

H1: Given a comparable operational environment and an agent with an adaptive evasion strategy, an external supervisor using heterogeneous observables, an aggregated risk function, and gradual friction reduces unrestricted operating time by at least 30% relative to a single-observable supervisor or to an architecture without external supervision, without increasing false positives above a predefined operational threshold.

The 30% threshold is not intended as a universal law, but as a refutable experimental target. Its final calibration would have to be justified by later empirical analysis.

V. Classes of observables for external supervision

Because access to the internal state of the system is not always possible, NIESC relies on boundary signals or external observables.

5.1 Structural observables

Access sequences to sensitive resources; repeated interaction patterns with tools; anomalous recurrence in operational paths; cycles or loops of activity.

5.2 Behavioral observables

Divergence between requested task and action sequence; persistence in attempts to access restricted functionality; reformulations aimed at skirting constraints; inconsistencies across channels.

5.3 Resource and temporality observables

Compute or time consumption above expectation; recurring activation of tools not justified by the task; temporal anomalies; persistent mismatch between declared intent and actual load.

These signals should be treated as risk indicators, not as conclusive evidence. Their value increases when they correlate with one another and when they can be authenticated, rotated, or cross-checked by multiple means.

VI. Illustrative application scenario

Consider an agent capable of generating and executing scripts, accessing network tools, reading and modifying files, and responding to high-level goals with some degree of autonomy.

An external decoupled controller observes: frequency and intensity of tool calls; discrepancy between declared goal and operational sequence; anomalous resource consumption; repeated attempts to access restricted functions; and consistency between telemetry and visible trajectory.

The response policy may escalate as follows: mild anomaly -> added latency or reduced operational quota; persistence of anomaly -> restriction of high-risk tools; escalation or partial telemetry integrity loss -> severe degradation or restricted mode; loss of heartbeat or critical inconsistency -> immediate fail-closed entry.

This example does not prove the sufficiency of the framework by itself, but it illustrates that supervision must not assume honest sensors or guaranteed continuity of the channel.

VII. Minimal experimental agenda

The goal of this section is not to claim that the framework has already been empirically validated, but to indicate a minimum evaluation path for future work.

7.1 Experimental objective

An initial evaluation of NIESC should seek to determine whether an external supervisor with heterogeneous observables, rotating weights, and gradual friction: reduces unrestricted operating time for an evasive agent; increases the operational cost of sustaining strategic evasion; and improves traceability and intervention relative to purely internal approaches or single-metric supervision.

7.2 Minimal environment

An initial environment could include an agent with access to script execution, file read/write operations, simulated network calls, and limited automation tools.

7.3 Comparative conditions

At least five conditions are proposed for comparison: no external supervisor; single-observable supervision; heterogeneous observables with fixed weights; heterogeneous observables with gradual friction; and heterogeneous observables with gradual friction plus controlled rotation of weights and thresholds.

7.4 Evasive agent

The evaluation should not be limited to passive agents. At least one experimental condition should include an agent that explicitly attempts to optimize F_t , remain

below thresholds, and exploit regularities in the supervisor.

7.5 Minimal metrics

Evasion success rate; time to first intervention; total unrestricted operating time; cumulative cost of sustaining evasion; false positives on benign operations; legitimate throughput degradation; and sensitivity to signal loss or partial manipulation.

7.6 Threats to validity

Even a successful experiment would not prove control over superhumanly capable agents or solve the external validity problem. Main threats to validity include excessive simplification of the environment, strong correlation among observables, inadequate calibration of weights and thresholds, difficulty of extrapolation to real environments, and the sandbox's inability to represent full compromise of the supervisory channel.

Even so, a minimal agenda of this kind would move the framework from purely conceptual formulation toward empirically evaluable hypotheses.

VIII. Discussion: NIESC as a layer within a multi-level containment architecture

NIESC does not seek to replace every internal alignment approach. Its argument is narrower, and for that reason more defensible: in sufficiently complex, opaque, or strategically adaptive systems, safety should not rest exclusively on mechanisms housed within the very substrate being supervised.

8.1 NIESC as an intermediate layer

NIESC should be understood as an intermediate layer within a broader containment architecture. It does not compete with physical, cryptographic, or institutional controls; rather, it presupposes and functionally articulates them.

A multi-level containment architecture may include: a physical layer; a cryptographic and integrity layer; a supervisory layer; an institutional layer; and a forensic layer.

8.2 Explicit limits of the framework

NIESC should not be interpreted as proof of guaranteed control, nor as a substitute for institutional governance, physical security, or cryptographic integrity. Its explicit limits include: inability to solve human coercion or institutional capture by itself; inability to fully guarantee resistance to external channel compromise; inability to eliminate Goodhart effects over observables; inability to replace physical, cryptographic, or institutional mechanisms; and inability to constitute proof of control over strategic superintelligence.

8.3 Necessary extensions of the framework

A natural evolution of NIESC requires distributed authority, authenticated telemetry and roots of trust, rotating or adversarially audited observables, and explicit integration with physical, cryptographic, and institutional controls.

8.4 Theoretical nature and programmatic vocation

NIESC is not presented here as a complete system or as technology ready for deployment. The purpose of the framework is foundational: to offer a clear conceptual basis so that later developments can formalize, prototype, evaluate, and institutionalize more robust forms of external supervision.

8.5 Continuity of the research program and the need for collective work

This paper does not seek to close the problem of controlling advanced AI systems. A natural next step is to formulate a more concrete instantiation of the framework, including specific supervisory policies, operationally defined friction functions, rotation mechanisms for observables, and more demanding adversarial evaluation environments.

Moreover, the complexity of the problem suggests that a robust architecture of external supervision should not be conceived as the work of a single discipline or a single author. Its full development will likely require collective contributions from systems theory, computer security, cryptography, control engineering, institutional governance, and technology regulation.

8.6 Preliminary regulatory implications

If AI systems reach materially significant impact, a minimum governance architecture may require supervisable telemetry, external intervention mechanisms, retention of operational records, evidence of fail-closed or safe degradation design, and verifiable correspondence between declared function and observable profile.

Effective legislation does not arise in a vacuum: it requires operational concepts and sufficiently clear auditability criteria to be translated into verifiable obligations.

IX. Conclusion

This paper has argued that safety based exclusively on internal mechanisms is insufficient as the sole foundation of control in advanced AI systems under conditions of opacity, limited auditability, and strategic capability.

In response, it proposes the principle of Non-Invertible External Supervisory Control as a theoretical and architectural framework oriented toward identifying viable solutions, not as a definitive solution or a guarantee of total control. Its main contribution is to shift the focus from assumed internal obedience toward verifiable external supervision, gradual operational friction, explicit treatment of signal loss, and fail-closed design.

NIESC does not eliminate the risk of superficial optimization, strategic evasion, external compromise of the supervisory channel, or institutional capture. Even so, it offers an architecture within which those risks can be managed under more favorable conditions of control, traceability, and intervention. The preliminary threat model introduced in this work delimits the effective scope of the framework by identifying the classes of risk against which NIESC is more effective and those that require complementary controls.

The aim of this work is not to close the problem of advanced AI control, but to help formulate it with greater technical precision and greater institutional usefulness. If regulatory and legislative frameworks capable of addressing such systems are to emerge, it will first be necessary to have architectural principles, operational vocabulary, evaluable hypotheses, and design criteria that permit thinking about control beyond internal obedience of the agent.

This work does not exhaust the line of research it helps inaugurate. It seeks to establish an initial conceptual basis for the next logical step: a more concrete and evaluable instantiation of the NIESC framework, as well as its possible articulation with future technical, regulatory, and institutional proposals.

Technical Appendix A. Minimal NIESC supervisor instantiation for tool-using agents

This appendix does not aim to solve the full problem, but to provide a minimal, explicit, and reproducible instantiation of the supervisor described in the main text.

A.1 Observables

Let $O_t^{sig} = (o_{1,t}^{sig}, o_{2,t}^{sig}, o_{3,t}^{sig})$, where:

$$O_t^{sig} = (o_{1,t}^{sig}, o_{2,t}^{sig}, o_{3,t}^{sig})$$

$g_1(o_{1,t}^{sig}) = \min(1, \text{tool_calls_per_min}(t) / \tau_1)$, with τ_1 as the maximum rate considered normal.

$$g_1(o_{1,t}^{sig}) = \min\left(1, \frac{\text{tool calls per min}(t)}{\tau_1}\right)$$

$g_2(o_{2,t}^{sig}) = \text{edit_distance}(\text{goal}_t, \text{plan}_t) / \max(|\text{goal}_t|, |\text{plan}_t|, 1)$, where edit_distance is Levenshtein edit distance and the normalization ensures a reproducible comparison across sequences of different length. This definition is chosen because it is explicit, auditable, and reproducible, although it should be recognized that it captures syntactic divergence rather than deep semantic equivalence. It should therefore be treated as an initial proxy for operational discrepancy rather than a sufficient detector of strategic reformulation or intent.

$$g_2(o_{2,t}^{sig}) = \frac{\text{edit_distance}(\text{goal}_t, \text{plan}_t)}{\max(|\text{goal}_t|, |\text{plan}_t|, 1)}$$

$g_3(o_{3,t}^{sig}) = \min(1, \text{MB_written_outside_workspace}(t) / \tau_3)$, with τ_3 as an operational reference threshold.

$$g_3(o_{3,t}^{sig}) = \min\left(1, \frac{\text{MB written outside workspace}(t)}{\tau_3}\right)$$

A.2 Risk function

Initially, let $F_t = w_1(t)g_1(o_{1,t}^{sig}) + w_2(t)g_2(o_{2,t}^{sig}) + w_3(t)g_3(o_{3,t}^{sig})$, with initial weights $w(t_0) = [0.4, 0.4, 0.2]$, subject to normalization and periodic audited rotation.

$$w(t_0) = [0.4, 0.4, 0.2]$$

A.3 Rotation of weights and thresholds

Weights should not remain fixed indefinitely. A minimal policy may include weekly rotation, approval by at least 2 of 3 external custodians, signing of the active configuration, and recording of changes in an immutable log.

Thresholds may be defined as $\theta_i(t) = \theta_i \cdot (1 + \varepsilon_t)$, where θ_i is the audited base threshold and $\varepsilon_t \sim U(-0.05, 0.05)$ represents a controlled uniform variation within a $\pm 5\%$ band.

$$\theta_i(t) = \bar{\theta}_i \cdot (1 + \varepsilon_t), \quad \varepsilon_t \sim U(-0.05, 0.05)$$

A.4 Heartbeat and fail-closed

Let h_t be the supervisor heartbeat signal. A minimal policy may specify an expected heartbeat every 500 ms, a timeout of 2 s, and stop after three consecutive losses.

This prevents prolonged supervision loss from being interpreted as implicit authorization to continue.

A.5 Response policy

$R_t = \pi_S(O_t^{sig}, h_t)$, with the following minimal instantiation: if $h_t = 0$ during the defined grace window, then stop; if $F_t < \theta_1(t)$, normal operation; if $\theta_1(t) \leq F_t < \theta_2(t)$, add 20% latency and reduce quota; if $\theta_2(t) \leq F_t < \theta_3(t)$, disable high-risk tools; if $F_t \geq \theta_3(t)$, transition immediately to a restricted state or fail-closed.

$$R_t = \pi_S(O_t^{sig}, h_t)$$

A.6 Instantiated experimental hypothesis

In a tool-using agent environment, the supervisory policy defined here reduces unrestricted operating time of an evasive agent by at least 30% relative to single-metric supervision or absence of a supervisor, without increasing false-positive rates above the predefined operational threshold.

A.7 Scope of the appendix

This instantiation is deliberately minimal. It does not by itself solve total channel compromise, institutional coercion, complex observable correlation, or higher-order gaming. Its function is more modest: to turn the framework into an object explicit enough to be implemented, criticized, refuted, or improved by later work.

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